WORKED SOLUTION

STEP I 2009, Question 3

Pure • Counting Real Roots

> **STEP I 2009, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/09-s1-q3

The key idea

Get rid of the root by raising both sides to the right power, but keep track of what that loses. A square root is never negative, so squaring is only reversible if we also insist that $x \geq 0$; a real cube root has no such restriction, because every real number has exactly one real cube root. After that, each count comes down to the roots of a quadratic, and the signs of those roots can be read off from their sum and product.

Solution

Part (i): the square root equation

Here $(a-x)^{\frac{1}{2}}$ means the non-negative square root, so any solution has $x \geq 0$, and squaring gives $x^2 = a - x$, that is $x^2 + x - a = 0$. Conversely, if $x \geq 0$ and $x^2 + x - a = 0$, then $a - x = x^2 \geq 0$ and $(a-x)^{\frac{1}{2}} = \sqrt{x^2} = |x| = x$. So

$$x = (a-x)^{\frac{1}{2}} \iff x^2 + x - a = 0 \text{ and } x \geq 0,$$

and we have to count the *non-negative* roots of the quadratic $x^2 + x - a = 0$. If its roots are α, β (when real), then

$$\alpha + \beta = -1, \qquad \alpha\beta = -a.$$

Case $a > 0$. The discriminant $1 + 4a$ is positive, so there are two real roots, and their product $-a$ is negative. So one root is positive and the other negative: exactly one solution.

Case $a = 0$. The quadratic is $x(x+1) = 0$, with roots 0 and -1. Only $x = 0$ is non-negative: exactly one solution.

Case $a < 0$. If $a < -\frac{1}{4}$ the discriminant is negative and there are no real roots at all. If $-\frac{1}{4} \leq a < 0$ the roots are real, their product $-a$ is positive, so they have the same sign, and their sum is -1, so both are negative. Either way there is no non-negative root, hence no solution.

So the equation has one real solution when $a \geq 0$ and none when $a < 0$, as required.

Part (i): the cube root equation

Every real number t has exactly one real cube root, and $x = t^{\frac{1}{3}} \iff x^3 = t$, with no sign condition. So the equation is equivalent to

$$x^3 - (1+a)x + a = 0.$$

The coefficients sum to $1 - (1+a) + a = 0$, so $x = 1$ is always a root, and dividing out,

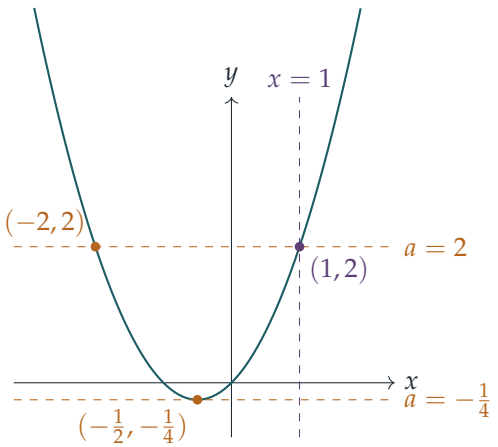
$$x^3 - (1+a)x + a = (x-1)(x^2 + x - a).$$

The same quadratic has appeared, and the solutions are $x = 1$ together with the real roots of $x^2 + x - a = 0$, now with no sign restriction.

> **Ask yourself** Could a root of the quadratic coincide with $x = 1$, or with each other?

Both happen, and they are exactly the cases that reduce the count. The quadratic has a repeated root when $1 + 4a = 0$, that is $a = -\frac{1}{4}$, the root being $-\frac{1}{2}$. And $x = 1$ is a root of the quadratic when $1 + 1 - a = 0$, that is $a = 2$; then $x^2 + x - 2 = (x-1)(x+2)$, so the roots are 1 and -2.

Writing the quadratic as $a = x^2 + x$, its real roots are where the horizontal line at height a meets the parabola $y = x^2 + x$, whose lowest point is $(-\frac{1}{2}, -\frac{1}{4})$. The extra solution $x = 1$ corresponds to the point $(1, 2)$ on the parabola.



Putting the cases together, the number of distinct real solutions is:

- $a < -\frac{1}{4}$: the quadratic has no real roots, so **one** solution, $x = 1$;
- $a = -\frac{1}{4}$: the solutions are 1 and $-\frac{1}{2}$, so **two**;
- $-\frac{1}{4} < a < 2$ or $a > 2$: two distinct roots of the quadratic, neither equal to 1, plus $x = 1$, so **three**;

- $a = 2$: the solutions are 1 and -2, so **two**.

Part (ii)

Exactly as in part (i), the square root is non-negative, so

$$x = (b+x)^{\frac{1}{2}} \iff x^2 - x - b = 0 \text{ and } x \geq 0,$$

(the converse holds because $x \geq 0$ and $b + x = x^2$ give $(b+x)^{\frac{1}{2}} = |x| = x$). Now the roots of $x^2 - x - b = 0$ have sum 1 and product $-b$, and the discriminant is $1 + 4b$.

Case $b < -\frac{1}{4}$. No real roots, so **no** solutions.

Case $b = -\frac{1}{4}$. The quadratic is $(x - \frac{1}{2})^2 = 0$, and $x = \frac{1}{2} \geq 0$, so **one** solution.

Case $-\frac{1}{4} < b < 0$. Two distinct real roots with positive product $-b$, so they have the same sign, and positive sum 1, so both are positive: **two** solutions.

Case $b = 0$. The quadratic is $x(x-1) = 0$, and both 0 and 1 are non-negative: **two** solutions. (Indeed $0 = \sqrt{0}$ and $1 = \sqrt{1}$.)

Case $b > 0$. Two real roots with negative product, so one is positive and one negative: **one** solution.

In summary: no solutions for $b < -\frac{1}{4}$; one for $b = -\frac{1}{4}$; two for $-\frac{1}{4} < b \leq 0$; one for $b > 0$.

Checks

- Graphically, part (ii) asks where the line $y = x$ meets the half-parabola $y = \sqrt{b+x}$, which starts at $(-b, 0)$. For $b = -\frac{1}{5}$ the solutions are $\frac{1}{2} \pm \frac{1}{2\sqrt{5}}$, about 0.276 and 0.724, and both satisfy $x = \sqrt{x - 0.2}$ to many decimal places.
- In the cube root equation with $a = 2$, both $x = 1$ and $x = -2$ work: $(3 \cdot (-2) - 2)^{\frac{1}{3}} = (-8)^{\frac{1}{3}} = -2$.
- A computer counted the roots of each equation on a fine grid for a range of values of a and b either side of every boundary ($-\frac{1}{4}$, 0, 2), and the counts agreed with the case analysis above.

Common slips

- Squaring and forgetting that $x \geq 0$: this would give two solutions in part (i) for every $a > -\frac{1}{4}$, which is wrong.
- Imposing $x \geq 0$ on the cube root equation as well. Cube roots of negative numbers are perfectly good real numbers.
- Missing the case $a = 2$, where the root $x = 1$ of the linear factor is also a root of the quadratic, so two of the three solutions merge.

Beyond STEP

The key point is that squaring is not a reversible operation on $\mathbb{R}$, while cubing is: $x \mapsto x^3$ is a bijection of $\mathbb{R}$, but $x \mapsto x^2$ is not, so "the" square root is only defined once we choose the non-negative branch. Thinking of equations like $x = f(x)$ as fixed points, and counting them as a parameter varies, is the start of bifurcation theory, where solutions are born in pairs (as at $a = -\frac{1}{4}$) or exchange places (as at $a = 2$). For a careful account of functions, inverses and continuity, see Abbott, *Understanding Analysis*.

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